线性代数 II

2026年 11月 2日.

2. 线性映射 —— 矩阵.

e.g. $V = \mathbb{R}[x]_{\le 2} = \mathrm{span}\{1, x, x^2\}$

$$\begin{aligned} D: V &\longrightarrow V \\ 1 &\longmapsto 0 \\ x &\longmapsto 1 \\ x^2 &\longmapsto 2x \end{aligned}$$

$$(D1,\ Dx,\ Dx^2)$$
$$\shortparallel$$
$$(\underline{0}\quad \underline{\underline{1}}\quad \underline{\underline{\underline{2x}}}) = (1\quad x\quad x^2)\begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 2 \\ 0 & 0 & 0 \end{pmatrix}$$

$$D\,(\text{Basis}) = (\text{Basis}) \cdot M$$

同一个 D, 在不同基下, 可能有不同的矩阵表示.

① $(D1\quad Dx\quad Dx^2) = \underbrace{(1\quad x\quad x^2)}_{}^{B_1}\begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 2 \\ 0 & 0 & 0 \end{pmatrix}$ M_1

② $(D1\quad D(1+x)\quad D(1+x)^2) = (1\quad 1+x\quad 1+x^2)\begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 2 \\ 0 & 0 & 5 \end{pmatrix}$

$$③\ (Dx\quad D1\quad Dx^2) = \underbrace{(x\quad 1\quad x^2)}_{B_3}\begin{pmatrix}0&0&2\\1&0&0\\0&0&0\end{pmatrix}\ M_3$$

基变换: $B_1 = B_3 \cdot ?$

$$⊛\quad \underbrace{(1\quad x\quad x^2)}_{B_1} = \underbrace{(x\quad 1\quad x^2)}_{B_3}\underbrace{\begin{pmatrix}0&1&0\\1&0&0\\0&0&1\end{pmatrix}}_{P}$$

$$①\quad DB_1 = B_1 M_1$$

$$③\quad DB_3 = B_3 M_3$$

$$⊛\quad B_1 = B_3 \cdot P$$

$$\Rightarrow D B_3 \cdot \overline{P} = B_3 \cdot P \cdot M_1$$

$$DB_3 = B_3 \cdot \underbrace{P \cdot M_1 \cdot P^{-1}}$$

$$\Rightarrow P M_1 P^{-1} = M_3$$

eg. $$\boxed{\frac{1}{1+\sqrt{2}+\sqrt{3}}} = a + b\sqrt{2} + c\sqrt{3} + \sqrt{6}$$

$$\parallel$$

$$\frac{1+\sqrt{2}-\sqrt{3}}{(1+\sqrt{2})^2-3} = \frac{1+\sqrt{2}-\sqrt{3}}{2\sqrt{2}} = \underbrace{\frac{1}{2}+\frac{\sqrt{2}}{4}-\frac{\sqrt{6}}{4}}$$

$$\textcircled{?}\quad \frac{1}{1+\sqrt{2}+\sqrt{3}+\sqrt{6}+\sqrt[3]{2}} = a + b\sqrt{2} + \cdots\ c\sqrt[3]{2}$$

$\textcircled{\sqrt{2}}$ $\textcircled{i}$

如何定义 $\underbrace{\sqrt{2}}_{x^2=2\text{ 的解}}$ (不依赖于拆字)、

$$\sqrt{2}: \quad \begin{array}{l} 1 \longmapsto \sqrt{2} \\ \sqrt{2} \longmapsto 2 \end{array}$$

$$\sqrt{2} \cdot (1 \quad \sqrt{2}) = (1 \quad \sqrt{2}) \begin{pmatrix} 0 & 2 \\ 1 & 0 \end{pmatrix}$$

$$\shortparallel$$

$$(\sqrt{2},\ 2)$$

$$2 = \sqrt{2}^2 \qquad \begin{pmatrix} 0 & 2 \\ 1 & 0 \end{pmatrix}^2 = \underset{2I}{\begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix}}$$

$\mathbb{Q}^4$

eg、$(1+\sqrt{2}+\sqrt{3}) \cdot (1, \sqrt{2}, \sqrt{3}, \sqrt{6})$

$$\shortparallel$$

$$(1+\sqrt{2}+\sqrt{3},\ 2+\sqrt{2}+\sqrt{6},\ 3+\sqrt{3}+\sqrt{6},\ 3\sqrt{2}+2\sqrt{3}+\sqrt{6})$$

$$= (1, \sqrt{2}, \sqrt{3}, \sqrt{6}) \cdot \begin{pmatrix} 1 & 2 & 3 & 0 \\ 1 & 1 & 0 & 3 \\ 1 & 0 & 1 & 2 \\ 0 & 1 & 1 & 1 \end{pmatrix}$$

$$\frac{1}{1+\sqrt{2}+\sqrt{3}} \longrightarrow \begin{pmatrix} 1 & 2 & 3 & 0 \\ 1 & 1 & 0 & 3 \\ 1 & 0 & 1 & 2 \\ 0 & 1 & 1 & 1 \end{pmatrix}^{-1}$$

问题. 证明 $\frac{1}{(1+\sqrt{2}+\sqrt{3})}$ 形如 $a+b\sqrt{2}+c\sqrt{3}+d\sqrt{6}$ $\quad a,b,c,d\in\mathbb{Q}$

证明. 由 $\left[(1+\sqrt{2}+\sqrt{3})\cdot : \begin{array}{ccc} \mathbb{Q}^4 & \longrightarrow & \mathbb{Q}^4 \\ x & \longmapsto & (1+\sqrt{2}+\sqrt{3})x \end{array}\right]$

是单射 知其为满射

所以 $1\in\mathbb{Q}^4$ 的原像即为所求 #

3 线性无关的技巧、

e.g. $\{1, x, x^2, x^3, \cdots, x^n\}$ 在 $\mathbb{R}$ 上线性无关

e.g. $\{1, \sqrt{2}, \sqrt{3}, \sqrt{5}, \sqrt{6}, \sqrt{7}, \sqrt{15}\}$ 在 $\mathbb{Q}$ 上线性无关

e.g. $\{\underline{\sin x, \sin^2 x, \sin^3 x}\}$ 在～上线性无关

$\{\sin x, \sin 2x \ \sin 3x\}$

$\boxed{\{e^x, e^{2x}, e^{3x}, e^{4x}\}}$

证明. $\underline{ae^x+be^{2x}+ce^{3x}+de^{4x}}=0$

下证 $a=b=c=d=0$

$$D^{k}\downarrow$$

$$ae^{x}+2^{k}be^{2x}+3^{k}ce^{3x}+4^{k}de^{4x}=0$$

$$\underbrace{\begin{pmatrix}1&1&1&1\\1&2&3&4\\1&4&9&16\\1&8&27&64\end{pmatrix}}_{vdm}\begin{pmatrix}ae^{x}\\be^{2x}\\ce^{3x}\\de^{4x}\end{pmatrix}=0$$

eg、设 $U_1, U_2, \cdots, U_m$ 是 $\mathbb{R}^n$ 的线性真子空间.

则 $U_1 \cup U_2 \cup \cdots \cup U_m \neq \mathbb{R}^n$

pf. 对任意 $x \in \mathbb{R}$，取 $v_x = (1, x, x^2, \cdots, x^{n-1})$

则 $\{v_x\}_{x\in\mathbb{R}} \subseteq \mathbb{R}^n$

$\{v_x\}_{x\in\mathbb{R}}$ 中任取 n 个都是线性无关组.

$\Rightarrow$ 每个 U_i 最多只能包含 $(n-1)$ 个

$\{v_x\}_{x\in\mathbb{R}}$ 中的元素. #

2. rank 相抵标准形

$$\begin{tikzcd} X \arrow[rr, "f"] \arrow[dr, "满"'] & & Y \\ & 像 \arrow[ur, "单"'] & \end{tikzcd}$$

$f:$

$$f\begin{pmatrix} x \\ y \\ z \end{pmatrix} \longmapsto \begin{pmatrix} x+2y \\ 2x+4y \end{pmatrix}$$

$$\begin{tikzcd} 3 \quad \mathbb{R}^3 \arrow[rr, "\begin{pmatrix} 1 & 2 & 0 \\ 2 & 4 & 0 \end{pmatrix}"] \arrow[dr, "满"'] & & \mathbb{R}^2 \quad 2 \\ & \text{列空间} \; 1 \arrow[ur, "单"'] & \end{tikzcd}$$

矩阵 ↓ 映射.

映射的秩 = 列空间维数 = 矩阵的秩.

$$\begin{pmatrix} \overset{f(e_1)}{f\begin{pmatrix}1\\0\\0\end{pmatrix}} & \overset{f(e_2)}{f\begin{pmatrix}0\\1\\0\end{pmatrix}} & \overset{f(e_3)}{f\begin{pmatrix}0\\0\\1\end{pmatrix}} \end{pmatrix}$$

$$\parallel$$

$$\begin{pmatrix} \begin{pmatrix}1\\2\end{pmatrix} & \begin{pmatrix}2\\4\end{pmatrix} & \begin{pmatrix}0\\0\end{pmatrix} \end{pmatrix} = \begin{pmatrix} \overset{e_1'}{\begin{pmatrix}1\\0\end{pmatrix}} & \overset{e_2'}{\begin{pmatrix}0\\1\end{pmatrix}} \end{pmatrix} \cdot \underrightarrow{\begin{pmatrix} 1 & 2 & 0 \\ 2 & 4 & 0 \end{pmatrix}}$$

映射 ↓ 矩阵.

SLOGAN. 矩阵A = 左乘A 在标准基下的矩阵表示

$$\begin{tikzcd}\mathbb{F}^{m\times n} \arrow[r, bend left, "左乘"] & \{\text{线性映射 } \mathbb{F}^n \to \mathbb{F}^m\} \arrow[l, bend left, "\text{标准基下的矩阵表示}"]\end{tikzcd}$$

列线性无关 ⟷ 单射.

列满秩 ⟷ 满射.

$N(A\cdot B)=0$ ⟷ $\boxed{g\circ f \text{ 单} \Rightarrow f \text{ 单}}$

$\Downarrow$

$N(B)=0$

$M = C\cdot R$ ⟷ $f =$ 单 ∘ 满

其中 $N(C)=0$ 且 $N(R^T)=0$

$$\begin{pmatrix}1&2&0\\2&4&0\end{pmatrix} = \underbrace{\begin{pmatrix}1\\2\end{pmatrix}}\underline{\begin{pmatrix}1&2&0\end{pmatrix}}$$

$$\begin{tikzcd}\mathbb{R}^3 \arrow[rr, "\begin{pmatrix}1&2&0\\2&4&0\end{pmatrix}\cdot"] \arrow[dr] & & \mathbb{R}^2 \\ & \mathrm{span}\left\{\begin{pmatrix}1\\2\end{pmatrix}\right\} \arrow[ur] & \end{tikzcd}$$

$$\underbrace{\begin{pmatrix}1&0\\2&1\end{pmatrix}\cdot\begin{pmatrix}1\\0\end{pmatrix}}\ \boxed{\begin{pmatrix}1&0&0\end{pmatrix}}\cdot\underbrace{\begin{pmatrix}1&2&0\\0&1&0\\0&0&1\end{pmatrix}}$$

$\begin{pmatrix}1&0&0\\0&0&0\end{pmatrix}$

$\begin{pmatrix}1\\0\\0\end{pmatrix}$ $\begin{pmatrix}2\\-1\\0\end{pmatrix}$ $\begin{pmatrix}0\\0\\1\end{pmatrix}$ $\longrightarrow \begin{pmatrix}0\\0\end{pmatrix}$

$$M = P^{-1}\cdot\begin{pmatrix}I_r&0\\0&0\end{pmatrix}\cdot Q,$$

$(r = \operatorname{rank} M)$

$$\begin{tikzcd}\mathbb{R}^3 \arrow[r, "\begin{pmatrix}1&2&0\\2&4&0\end{pmatrix}\cdot"] \arrow[d, "Q\cdot"'] \arrow[dr, phantom, "\circlearrowleft"] & \mathbb{R}^2 \arrow[d, "P\cdot"] \\ \mathbb{R}^3 \arrow[r, "\begin{pmatrix}1&0&0\\0&0&0\end{pmatrix}\cdot"'] & \mathbb{R}^2\end{tikzcd}$$

$$\begin{tikzcd}\begin{pmatrix}1\\2\end{pmatrix} \arrow[d] & \begin{pmatrix}0\\1\end{pmatrix} \arrow[d] \\ \begin{pmatrix}1\\0\end{pmatrix} & \begin{pmatrix}0\\1\end{pmatrix}\end{tikzcd}$$

ex. $P, Q \in \mathbb{R}^{m\times n}$. 则以下等价.

$P^{-1} = \begin{pmatrix} 1 & 0 \\ 2 & 1 \end{pmatrix}$

$Q^{-1} = \begin{pmatrix} 1 & 2 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$

① $\operatorname{rank}(P+Q) = \operatorname{rank} P + \operatorname{rank} Q$

② $\exists$ 可逆 X, Y s.t. $XPY = \begin{pmatrix} I & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$

$XQY = \begin{pmatrix} 0 & 0 & 0 \\ 0 & I & 0 \\ 0 & 0 & 0 \end{pmatrix}$

Pf. Hint consider C·R.

ex. 酉/正交三角化.

Schur $X \in \mathbb{C}^{n\times n}$, $\exists\, U$ 酉 s.t. $U^H X U = \begin{pmatrix} * & * & * \\ 0 & * & * \\ & & \ddots \\ & & & * \end{pmatrix}$

实? $X \in \mathbb{R}^{n\times n}$ $\exists\, Q$ 正交 s.t.

$$Q^T \cdot X \cdot Q = \begin{pmatrix} * & * & & & & \\ * & * & & & \circledast & \\ & & * & & & \\ & & & * & & \\ & & & & * & \\ & & & & * & * \end{pmatrix}$$

ex. $M \in \mathbb{R}^{n\times n}$.

则 $\exists\, Q$ 正交 s.t. $Q^T \cdot M \cdot Q = \begin{pmatrix} R & S \\ 0 & 0 \end{pmatrix}$

行线性无关.

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eg. $A \in \mathbb{R}^{n\times n}$, $A^2 = A\cdot A^T$. w/ $A = A^T$ ✓

$A \in \mathbb{R}^{n\times n}$. $A^3 = AA^TA$, w/ $A = A^T$

pf. $A = Q\cdot\begin{pmatrix} R & S \\ 0 & 0 \end{pmatrix} Q^T$

$$A^2 = Q\cdot\begin{pmatrix} \boxed{R^2} & RS \\ 0 & 0 \end{pmatrix} Q^T$$

$$A\cdot A^T = Q\cdot\begin{pmatrix} \boxed{R\cdot R^T + S\cdot S^T} & 0 \\ 0 & 0 \end{pmatrix} Q^T$$

$\Rightarrow$ $\det R^2 = (\det R)^2$

$\det(R\cdot R^T + S\cdot S^T) = \det (R\ S)\cdot(R\ S)^T$

$\Rightarrow$ $S = 0$.